

Bases (bay-seez) (plural of 'basis')

Defⁿ V : F -vector space
 $X \subseteq V$: subset

X is a basis for V

if $\forall v \in V$, $\exists$ a unique linear
combination

$$\sum_{i=1}^m a_i v_i = v$$

Where $a_i \in F$, $v_i \in X$

Rmk ① This is equivalent to
saying:

$$(a) \text{ Span}(X) = V$$

(b) X is linearly independent

(i.e. there is no redundancy)

Examples:

$$(1) \quad V = \mathbb{F}^3$$

$$X = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Then

$$\begin{bmatrix} \underline{c_1} \\ \underline{c_2} \\ \underline{c_3} \end{bmatrix} = \underline{c_1} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \underline{c_2} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + \underline{c_3} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Therefore, X is a basis.

(2) More generally,

$$V = F^n$$

$$X = \left\{ \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \dots, \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix} \right\}$$

n vectors

X is a basis for V

Defⁿ

$$e_i = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} \leftarrow i\text{-th row}$$

is the i -th standard basis vector
for F^n &

$$X = \{e_1, e_2, \dots, e_n\}:$$

standard
basis

Example (3)

$$V = \text{Poly}(F)$$

$$X = \{1, x, x^2, x^3, \dots\}$$

Then X is a basis for V

$$Y = \{1, (x-1), (x-1)^2, \dots\}$$

Then Y is also a basis for V

Example (4)

$$V = \text{Poly}_{\leq 4}(V)$$

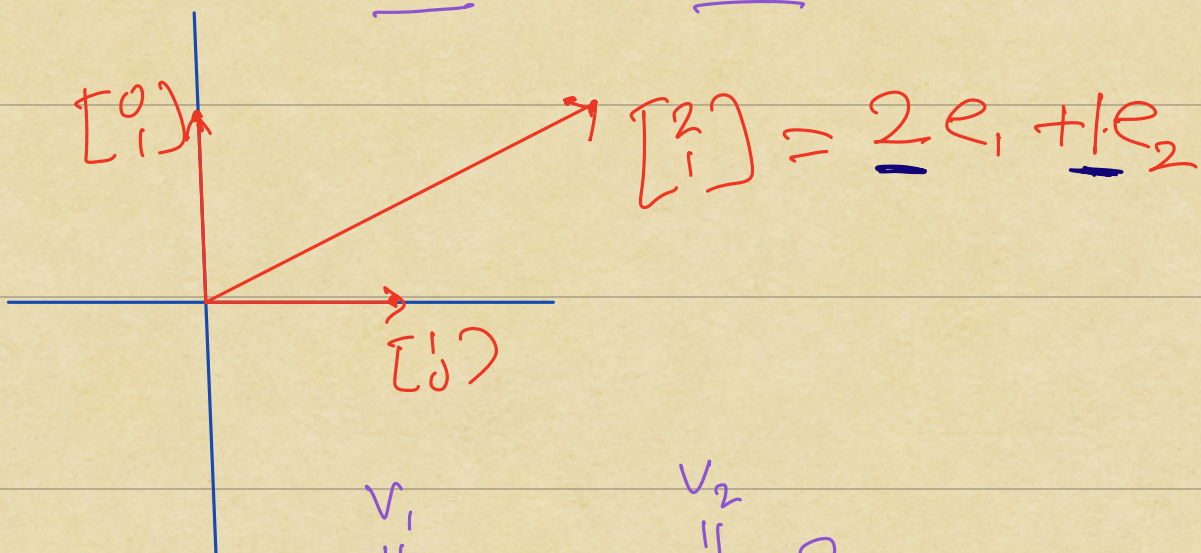
$$\{1, x, x^2, x^3, x^4\} : \text{basis}$$

$$\{1, (x-1), (x-1)^2, (x-1)^3, (x-1)^4\}$$

Example (5) $F = \mathbb{R}$

$$V = \mathbb{R}^2$$

$$X = \left\{ \underbrace{\begin{bmatrix} 1 \\ 0 \end{bmatrix}}_{e_1}, \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{e_2} \right\}$$



$$Y = \left\{ \underbrace{\begin{bmatrix} 1 \\ 0 \end{bmatrix}}_{v_1}, \underbrace{\begin{bmatrix} 2 \\ 1 \end{bmatrix}}_{v_2} \right\}$$

This is also a basis

(Linear independence): v_2 is not a scalar multiple of v_1

(Spans $\mathbb{R}^2$):

Need to show that

$$\text{Span}(\{v_1, v_2\}) = \mathbb{R}^2$$

i.e. Every $\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \in \mathbb{R}^2$ is

in $\text{Span}(\{v_1, v_2\})$

Rmk: If $\boxed{W} \in \text{Span}(\{v_1, v_2, \dots, \boxed{v_m}\})$

then

$$\begin{aligned} & \text{Span}(\{v_1, v_2, \dots, v_{m-1}, \boxed{W}\}) \\ & \subseteq \text{Span}(\{v_1, v_2, \dots, v_{m-1}, v_m\}) \end{aligned}$$

Pf:

$$W = c_1 v_1 + c_2 v_2 + \dots + c_m v_m$$

$$\text{If } v = d_1 v_1 + d_2 v_2 + \dots + d_{m-1} v_{m-1} + d_m W$$

then v is also a linear

Combination of $\{v_1, v_2, \dots, v_{m-1}, v_m\}$

Replace w with a linear combo
of $v_1, \dots, v_m$:

$$\begin{aligned} v &= d_1 v_1 + \dots + d_{m-1} v_{m-1} + d_m (c_1 v_1 + \dots + c_m v_m) \\ &= (d_1 + d_m c_1) v_1 + \dots + (d_{m-1} + d_m c_{m-1}) v_{m-1} \\ &\quad + d_m v_m \end{aligned}$$

Back to example (5)

$$\begin{array}{ccc} \text{Span}(\{v_1, v_2\}) & \stackrel{??}{=} & \mathbb{R}^2 \\ \parallel & & \parallel \\ \underline{\begin{bmatrix} 1 \\ 0 \end{bmatrix}} & \underline{\begin{bmatrix} 2 \\ 1 \end{bmatrix}} & \text{Span}(\{e_1, e_2\}) \end{array}$$

All we need is:

$$\text{Span}(\{e_1, \boxed{e_2}\}) \subseteq \text{Span}(\{v_1, v_2\})$$

$$\parallel$$
$$\text{Span}(v_1, v_2)$$

$$e_2 = v_2 - 2v_1$$

$$= \begin{bmatrix} 2 \\ 1 \end{bmatrix} - 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Theorem

V : F -vector space

(1) There exists a basis for V

(2) Every basis for V has the same number

of elements.

Defⁿ • A F -vector space V is **finite dimensional** if it admits a basis with finitely many elements.

(In this case)

• The **dimension** of V

is the size of any basis for V .
the number of elements

Notation: $\dim V$
or $\dim(V)$

Rmk: This is well-defined
by (2) of theorem